PECH Codebase Summary

Repository Overview

This repository contains the **complete strategic planning, business operations, and technical blueprint** for **PECH Group Holdings Ltd** — building Africa's first vertically-integrated Infrastructure Operating System and Commerce Operating System. It combines IoT hardware, AI-native software platforms, and a multi-vertical ecosystem across 10+ business domains.

What this repository IS: - Complete business strategy, financial projections, and market analysis - AI/ML strategy with 50+ model catalog, architecture guides, and hardware planning - Legal templates (contracts, agreements, policies — Nigerian law compliant) - Operational forms (HR, sales, finance, inventory — 25+ forms) - Brand guidelines, design system, and visual assets - Investor proposal documents (financial plan, executive summary, pitch materials) - Document generation scripts (Python, Bash)

What this repository is NOT (yet): - No production application code (no APIs, backends, frontends) - No database schemas or migrations - No IoT firmware or embedded code - No AI/ML model training code - No DevOps pipelines (no Terraform, Kubernetes configs, CI/CD)

Directory Structure

```
Pech-IOT-and-SMART-SYSTEMS-PLATFORM-FOR-AFRICA/
│
├── README.md                                          # Project
overview & navigation
├── CLAUDE.md                                          # AI assistant
guidelines
├── CODEBASE_SUMMARY.md                                # This file
│
├── ai_strategy/                                       # [36 files] AI
ecosystem docs, diagrams & PDFs
├── business_documents/                                # [76 files]
Operational forms (MD + DOCX + PDF)
├── contracts/                                         # [36 files]
Legal templates (MD + DOCX + PDF)
```

```
├── brand_templates/                                # [37 files]
Brand collateral (MD + DOCX + PDF)
├── employment_xlsx/                                # [40 files]
Excel versions (39 XLSX + README)
├── design_assets/                                  # [16 files]
Logos, graphics, templates
├── design_system/                                  # [2 files]
CSS design system + HTML components
├── conversations/                                  # Strategic
planning conversations
│   ├── PECH_STRATEGIC_VISION_COMPLETE.md/docx/pdf   #   Unified
reference document
│   ├── PECH_CLAUDE_INDEPENDENT_ANALYSIS.md/docx/pdf #   Claude's
independent analysis
│   ├── 01-The-Idea-to-Build.md/docx/pdf             #   Formatted
conversation 1
│   ├── 02-Ecosystem-Understanding.md/docx/pdf       #   Formatted
conversation 2
│   ├── 03-AI-Integrations-and-Adoptions.md/docx/pdf #   Formatted
conversation 3
│   ├── originals/                                   #   Original
unformatted transcripts
│   └── images/                                      #
Architecture diagrams
├── proposal_images/                                # [26 files]
Infographics (SVG + PNG)
├── scripts/                                        # [6 files]
Document generation scripts
├── .claude/commands/                               # [1 file]
Slash command templates
│
├── PECH_GROUP_FINANCIAL_PROPOSAL_250M_NAIRA.md/docx/pdf  # Full 24-
month budget
├── PECH_INVESTOR_VERSION.md/docx/pdf                     #
Investor-focused proposal
├── PECH_PROPOSAL_EXECUTIVE_SUMMARY.md/docx/pdf           # One-page
executive summary
├── PECH_CANDIDATE_APPLICATION_FORM.docx/xlsx             # Hiring
form
│
├── proposal_style.css                              # Proposal PDF
styling
├── the Idea to build                               # Original
ChatGPT strategy transcript
├── AI integrations and adoptions in our ecosystems  # AI
integration planning
├── must read, what to understand about our ecosystem # Ecosystem
context document
└── Review Claude's plan                             # Plan review
notes
```

Detailed Directory Breakdown

`ai_strategy/` — AI Ecosystem Documentation (36 files)

The core technical and strategic documentation for PECH's self-hosted AI infrastructure.

File	Size
`README.md`	2 KB

PECH_ECOSYSTEM_COMPREHENSIVE_GUIDE.md	39 KB
PECH_AI_MODEL_CATALOG.md	30 KB
PECH_OPEN_SOURCE_PLATFORM_STACK.md	31 KB
PECH_AI_HARDWARE_AND_SETUP_GUIDE.md	35 KB
PECH_AI_ARCHITECTURE_GUIDE.md	36 KB
CRM_ERP_INTEGRATION_RESEARCH.md	28 KB
NIGERIA_AFRICA_AUTOMATION_SMART_SYSTEMS_INDUSTRY_RESEARCH.md	22 KB
images/	—

Key license corrections from ChatGPT conversations: - Metabase (AGPL) → Apache Superset (Apache-2.0) - n8n (Sustainable Use License) → Apache Airflow (Apache-2.0) - YOLOv8 (AGPL) → RT-DETR PaddlePaddle (Apache-2.0) - Neo4j (GPL) → Apache AGE on PostgreSQL (Apache-2.0) - Nigerian salaries corrected from $2-4K/mo to ₦400K–₦1M/mo - Hardware costs include 25-35% Nigeria import duties + 7.5% VAT

`business_documents/` — Operational Forms (76 files)

25 Markdown templates + 26 DOCX exports + 25 PDF exports across 4 operational categories.

Category	Forms	Key Documents
HR & Operations	8	Application form, interview checklist, job handbook (37 roles), leave request, asset register, incident report, visitor log, staff ID
Sales & Orders	6	Quotation, proforma invoice, sales invoice, purchase order, work order, job completion report
Inventory & Logistics	5	Stock requisition, delivery note, goods received note, waybill, installation completion certificate
Finance & Payments	6	Payment receipt, payment voucher, petty cash voucher, expense report, credit note, debit note

Notable: `PECH_JOB_REQUIREMENTS_HANDBOOK.md` (72 KB) — all 37 roles across 5 departments with qualifications, compensation, and hiring timeline.

`contracts/` — Legal Templates (36 files)

12 Markdown templates + 12 DOCX exports + 12 PDF exports, all Nigerian law-compliant.

Contract	Purpose
Full-Time Employment Contract	Nigerian Labour Act compliant
Contract Worker Agreement	Independent contractor terms
Internship Agreement	3–6 month structured programs
Non-Disclosure Agreement	Confidentiality protection
Media & Public Communications Policy	Brand/communications governance
API Developer Agreement	Third-party API usage terms
Installer Agreement	Field installation partners
Guarantor Form	Employee guarantor documentation
Hackathon Rules & Guidelines	Developer event governance
Commission Marketer Agreement	Sales agent commission terms
Office Marketer Agreement	In-house marketing staff terms
Schedule B Templates	Contract appendix templates

`brand_templates/` — Brand Collateral (37 files)

13 Markdown templates + 12 DOCX exports + 12 PDF exports for consistent brand identity.

Template	Purpose
Brand Guidelines	Color palette (Sky Blue #00BFFF, Orange #F5A623), usage rules
Business Card	Staff business card layout
Letterhead	Official correspondence header
Invoice (Branded)	Customer-facing invoice design
Email Signature	Standard email signature format
Certificate	Completion/training certificates
Presentation	Slide deck template
Newsletter	Monthly/quarterly newsletter format
Report Cover	Technical/business report cover page
Social Media	Platform-specific post templates
Merchandise Guidelines	Branded merchandise specs
Signage	Office/event signage standards

`design_assets/` — Visual Assets (16 files)

```
design_assets/
├── logos/          # Primary, horizontal, white, icon-only variants
(SVG + PNG) + favicons
├── graphics/       # Color palette, gradient bar, pattern
background, watermark (SVG)
└── templates/      # Template-specific assets
```

`design_system/` — CSS Design System (2 files)

File	Size	Description
`pech_design_system.css`	19 KB	Complete CSS design system with brand variables, typography, components (buttons, cards, forms, tables), responsive breakpoints, animations
`pech_components.html`	13 KB	HTML component library and visual showcase — demonstrates all design system components

`proposal_images/` — Infographics (26 files)

13 SVG source files + 13 PNG exports for proposal documents:

- 15-year expansion blueprint
- Budget allocation breakdown
- Commerce OS architecture (6-layer ecosystem)
- Compliance timeline
- Customer journey map
- Ecosystem overview
- Market entry map (Nigeria → Pan-African)
- Phase roadmap (Crawl → Walk → Run)
- Product portfolio
- Revenue projections

- Risk matrix
- ROI analysis
- Technology stack overview

`employment_xlsx/` — Excel Spreadsheets (40 files)

39 branded Excel (.xlsx) files covering all PECH documents + 1 README:

Category	Count	Description
Contracts	9	Employment, contractor, NDA, installer, marketer agreements
HR Forms	5	Application form, interview checklist, leave request, staff ID, job handbook
Financial	10	Invoices, receipts, vouchers, expense reports, quotations, purchase orders
Operations	12	Delivery notes, waybills, stock requisitions, work orders, incident reports
HR Extended	1	Schedule B compensation tracker
Dashboards	2	HR Master Dashboard + Master Document Dashboard

All files feature PECH branding, dropdown validations, alternating row colors, and Google Sheets compatibility.

`scripts/` — Document Generation & Branding (6 files)

Script	Language	Purpose
`generate_application_form.py`	Python (35 KB)	Generates branded Excel (.xlsx) and Word (.docx) application forms with clickable checkboxes
`generate_employment_xlsx.py`	Python (62 KB)	Generates 15 employment contract/HR Excel files with PECH branding
`generate_business_xlsx.py`	Python (59 KB)	Generates 24 business/operational Excel files with PECH branding
`regenerate_proposals.py`	Python (13 KB)	Converts all 59 Markdown files → PDF + DOCX with PECH brand styling (headers, colored headings, gradient tables, branded footers)
`brand_docx.py`	Python (20 KB)	Post-processes DOCX files with PECH brand styling: colored dividers, branded headers, table formatting

`export_hr_docs.sh`	Bash (6 KB)	Batch exports HR Markdown documents to PDF/DOCX via Pandoc + texlive-xetex

`.claude/commands/` — AI Assistant Commands (1 file)

File	Purpose
`industry-research.md`	Slash command template for generating industry research reports — auto-adapts to any industry with 16+ mandatory sections

Financial Proposal Documents

Three versions of the ₦250M (~$160K USD) 24-month capital deployment plan:

Document	Audience	Size
`PECH_GROUP_FINANCIAL_PROPOSAL_250M_NAIRA` (.md/.docx/.pdf)	Full version — internal/board	76 KB (MD)
`PECH_INVESTOR_VERSION` (.md/.docx/.pdf)	Investor-focused with ROI emphasis	63 KB (MD)
`PECH_PROPOSAL_EXECUTIVE_SUMMARY` (.md/.docx/.pdf)	One-page overview for quick review	9 KB (MD)

Key financials: - **Total Budget:** ₦250M over 24 months - **Team:** 37 roles (24 FT + 5 contract + 8 intern) - **Phases:** Crawl (Months 1–6), Walk (7–12), Run (13–24) - **Hardware:** TRX50 workstation + RTX 4090 GPUs + solar backup - **Break-even:** Month 16–18 - **Revenue target:** ₦21.3M/month by Month 24

Original Strategy Documents

File	Description
`the-idea-to-build` (333 KB)	Original ChatGPT conversation transcript — 65 exchanges across 32 topics covering entire business vision, architecture, market strategy, and execution roadmap
`ai-integrations-and-adoptions-in-our-ecosystems` (49 KB)	AI integration planning and adoption strategy
`must-read-what-to-understand-about-our-ecosystem` (53 KB)	Ecosystem context and understanding document
`review-claudes-plan` (10 KB)	Review notes on Claude's proposed plans

ChatGPT Conversation Index (65 Exchanges)

The `the-idea-to-build` document contains **65 user prompts** across **32 major topics**:

#	Topic	Key Outputs
1	Business Model Foundation	IoT Product & Platform Company definition, open-source stack (ThingsBoard, EMQX)
2	Nigerian Market Strategy	5 business models, market segmentation, ESP32/STM32 stack
3–4	Architecture Diagrams & Branding	Infographic iterations, PECH Solar Cloud branding
5–6	Tuya Competitor Research	Tuya's African weaknesses, market control strategy
7	Comprehensive Solutions	10-part strategic framework, moat building
8	24-Month Execution	8-phase roadmap (foundation → Series A positioning)
9	Competitive Attack Plan	5 attack vectors against Tuya, risk matrix
10	5-Year Blueprint	Dominance blueprint 2026–2031, 8-layer architecture
11–12	AI + Manufacturing + Telecom	AI framework, OEM→local assembly path, telecom partnerships
13	AI Code + Government + Whitepaper	Production AI code structure, smart grid entry
14	10-Year Monopoly Strategy	Continental monopoly, pricing, fintech, smart city
15	Multi-Vertical Expansion	7-phase expansion, 10-year financial projections
16	15-Year Continental Blueprint	Infrastructure empire, 9-layer systems architecture
17–20	Infographic Iterations	Revenue projections, vertical additions, timeline corrections
21–22	Vertical Optimization	16-vertical structure (3 tiers), 4 new strategic verticals
23–24	Technical Architecture	7-layer enterprise architecture, 24-month engineering roadmap
25	Capital + Org + Regulatory	$3.2M 3-year capital model, NCC/NDPA/NEMSA compliance
26–27	Technical PRD + Licenses	Formal PRD, Nigerian regulatory/licensing matrix with costs
28	Open Source Repositories	11+ curated repos with stars, licenses, implementation roadmap
29–30	Cold Chain & Logistics	Nigeria's $2.3–3.3B food waste crisis, perishable logistics platform
31–32	Full Stack + CI/CD	24 production repos, CI/CD pipeline, project skeleton
33–36	Alibaba Model Analysis	Commerce segments, embedded ERP, Africa adaptation, fork vs build
37–39	YiwuGou Market Digitization	Nigerian market taxonomy, 12-month pilot plan, field agent playbook
40	Master Presentation	20-slide "Africa's Commerce Operating System"

41	Hardware-First ERP Model	Free ERP + hardware sales model (POS, printers, scanners)
42–65	Continued Deep Dives	Hardware catalog, manufacturing, IoT integration, pricing models, developer ecosystem

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Technology Stack Reference

AI Models (Self-Hosted, License-Safe Only)

Category	Models	License
Primary LLM	Qwen2.5 (0.5B–72B)	Apache-2.0
Secondary LLM	Phi-4, Mistral 7B	MIT, Apache-2.0
Code	Qwen2.5-Coder, DeepSeek-Coder-V2-Lite	Apache-2.0, MIT
Speech-to-Text	Whisper, Faster-Whisper	MIT
Text-to-Speech	Piper TTS, Kokoro TTS	MIT, Apache-2.0
Vision	Qwen2-VL, Qwen2.5-VL	Apache-2.0
OCR	PaddleOCR, Tesseract	Apache-2.0
Embeddings	MiniLM-L6, Nomic-Embed	MIT, Apache-2.0
Object Detection	RT-DETR	Apache-2.0
ML/Forecasting	XGBoost, LightGBM, Prophet	Apache-2.0, MIT

Platform Stack (All Open-Source)

Category	Platform	License
API Gateway	Apache APISIX	Apache-2.0
Identity	Keycloak	Apache-2.0
Commerce	Medusa	MIT
ERP	ERPNext	MIT
Fintech	Apache Fineract	Apache-2.0
IoT	ThingsBoard CE	Apache-2.0
Support	Chatwoot	MIT
Vector DB	Qdrant	Apache-2.0
BI/Analytics	Apache Superset	Apache-2.0
Workflow	Apache Airflow	Apache-2.0
Graph DB	Apache AGE on PostgreSQL	Apache-2.0
Model Serving	Ollama → vLLM	MIT, Apache-2.0
RAG	LangChain + LangGraph	MIT

Infrastructure

Category	Technologies
MCU/Hardware	ESP32, STM32, Quectel BG95 (NB-IoT/LTE-M)
Protocols	MQTT, HTTP/REST, CoAP, WebSockets, Modbus, LoRaWAN, NB-IoT, Matter

MQTT Brokers	EMQX (Apache-2.0), Mosquitto (EPL), NanoMQ (MIT)
Backend	Node.js, Go, Python (FastAPI)
Databases	PostgreSQL, TimescaleDB, Redis, Qdrant
Streaming	NATS (Phase 1) → Apache Kafka (Phase 3)
Frontend	React, React Native, Flutter
DevOps	Docker Compose → K3s → K8s, Terraform, GitHub Actions
Auth	Keycloak (OAuth2/JWT/RBAC)
Analytics	Apache Superset, Grafana
Payments	Paystack, Flutterwave, Mobile Money
Maps	Leaflet, MapLibre GL

Models to AVOID (License Issues)

Model	License	Issue
Llama 3/3.1/3.2	Meta License	Cannot rebrand
Gemma 2/3	Google Custom	Redistribution restrictions
YOLOv8	AGPL-3.0	Copyleft
Stable Diffusion 3+	Stability AI	Non-commercial

Architecture Overview

78 Microservices Across 12 Domains

PECH's platform architecture spans 5 layers:

1. **Sovereign Cloud Layer** — Self-hosted K8s on RTX 4090 GPUs in Lagos
2. **Core Platform** — APIs, data pipelines, model serving
3. **Open API Layer** — Webhooks, SDKs, marketplace via Apache APISIX
4. **AI Core** — RAG pipelines, agents, vision, speech, forecasting
5. **Vertical Applications** — Solar, marketplace, fintech, logistics, IoT, real estate

3-Phase Deployment

Phase	Timeline	Infrastructure	Key Milestone
Crawl	Months 1–6	TRX50 + Ollama, Docker Compose	MVP launch
Walk	Months 7–12	Add 2nd GPU + vLLM, K3s	Commercial launch
Run	Months 13–24	Co-locate + scale, full K8s	₦21.3M/month revenue

10 Business Verticals

1. Solar Intelligence
2. Marketplace Intelligence

3. Fintech Intelligence
4. Mapping (AMAP/PMAP)
5. Logistics Intelligence
6. IoT Intelligence
7. Developer Platform
8. Installer Ecosystem
9. ERP Intelligence
10. Real Estate Platform

File Statistics

Category	Count	Formats
Markdown Documents	67	.md
Word Documents	60	.docx
Excel Spreadsheets	40	.xlsx
PDF Documents	59	.pdf
SVG Graphics	30	.svg
PNG Images	26	.png
Python Scripts	5	.py
Bash Scripts	1	.sh
CSS Stylesheets	2	.css
HTML Files	1	.html
Strategy Transcripts	4	plaintext
Total	**295**	Mixed
Total Size	**~39 MB**	—

Licensing Strategy

- **Primary preference:** Apache-2.0 and MIT licenses (can rebrand, modify, close-source)
- **Acceptable:** EPL (with attribution), BSD
- **Avoid for core:** GPL/AGPL (copyleft — must open-source modifications)
- **Hard avoid:** Meta License, Google Custom License, AGPL-3.0, Stability AI License

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Nigeria-Specific Considerations

- **Import Duties:** 25–35% + 7.5% VAT on all hardware from China
- **Power:** UPS + solar backup required (Lagos power instability)
- **Data Protection:** NDPA 2023 compliance mandatory
- **Mobile-First:** 70%+ users on mobile, USSD fallback for feature phones
- **Co-location:** MainOne, Rack Centre, 21st Century (Lagos options)
- **Regulatory:** NCC Type Approval, NEMSA, NERC, CBN, SONCAP, Trademarks

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Summary

PECH Group Holdings Ltd is building **Africa's Infrastructure Operating System** — a vertically-integrated platform combining:

1. **IoT Hardware** — Smart meters, sensors, gateways, CCTV, POS devices
2. **AI-Native Platform** — 50+ self-hosted models on RTX 4090 GPUs, zero cloud dependency
3. **Commerce OS** — Alibaba-inspired 6-layer ecosystem (ERP → Marketplace → Payments → Logistics → Credit → Data)
4. **10 Business Verticals** — Solar, marketplace, fintech, mapping, logistics, IoT, developer, installer, ERP, real estate
5. **Open API Economy** — White-label, SDKs, webhooks via Apache APISIX gateway
6. **37-Person Team** — 24 FT + 5 contract + 8 intern with Nigerian salary ranges

Target: Nigeria first → West Africa → East & Southern Africa → Pan-African dominance over 15 years.

Current Status: Documentation and strategic planning complete. Phase 1 (Crawl) infrastructure deployment underway. No production application code exists yet — this repository serves as the comprehensive blueprint for engineering execution.

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